

# Mathematics-I (MAL1010)

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# What is a Set?

- A **set** is a collection of distinct well-defined objects.
- Elements are enclosed in curly braces:  $\{ \}$ .
- Example:  $A = \{1, 2, 3\}$
- If  $x$  is an element of  $A$ , we write  $x \in A$ .

# Types of Sets

- **Finite set:** Set with a limited number of elements.
- **Infinite set:** Set with infinite elements, e.g.,  $\mathbb{N} = \{1, 2, 3, \dots\}$
- **Empty set:**  $\emptyset$  or  $\{\}$ , a set with no elements.
- **Subset:**  $A \subseteq B$  means every element of  $A$  is in  $B$ .

# Set Operations

- **Union:**  $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- **Intersection:**  $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- **Difference:**  $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
- **Complement:**  $A^c = \text{all elements not in } A$

# Cartesian Product of Sets

- Given sets  $A$  and  $B$ , the Cartesian product is:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

- Example: If  $A = \{1, 2\}$  and  $B = \{x, y\}$ , then

$$A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$$

# Binary Operations

- A **binary operation** on a set  $S$  is a function:

$$* : S \times S \rightarrow S$$

- Takes two elements from  $S$  and returns one in  $S$ .
- Examples on  $\mathbb{R}$ :
  - Addition:  $a + b$
  - Multiplication:  $a \cdot b$
  - Subtraction:  $a - b$

# Properties of Binary Operations

Let  $*$  be a binary operation on set  $S$ .

- **Commutative:**  $a * b = b * a$
- **Associative:**  $(a * b) * c = a * (b * c)$
- **Identity element:** There exists  $e \in S$  such that  $a * e = e * a = a$
- **Inverse element:** For each  $a \in S$ , there exists  $b \in S$  such that  $a * b = e$

# Building Up the Number System

## Natural Numbers ( $\mathbb{N}$ )

- Counting numbers:  $1, 2, 3, \dots$

## Integers ( $\mathbb{Z}$ )

- Includes negative numbers and 0:  $\{\dots, -2, -1, 0, 1, 2, \dots\}$

## Rational Numbers ( $\mathbb{Q}$ )

- Numbers of the form  $\frac{m}{n}$  where  $m, n \in \mathbb{Z}, n \neq 0$



# The Number Line

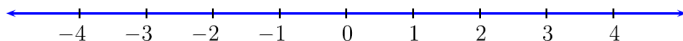


Fig. 1.1. The number line.

Figure:

- Integers can be placed on a line using a base point (0) and a unit distance.
- Rational numbers fill the line densely, but gaps remain.
- Example:  $\sqrt{2}$  lies between 1 and 2 but is not rational.
- These gaps lead to the concept of **irrational numbers**.

## Real Numbers

Rational and irrational numbers together form the set  $\mathbb{R}$ , called the **real numbers**.

# Order on Real Numbers

- The number line gives a natural **ordering** of real numbers.
- Numbers to the right of 0 are **positive**.
- Numbers to the left of 0 are **negative**.

## Example

$3 > 1$  means 3 is to the right of 1 on the number line.

# Types of Properties

The set of real numbers  $\mathbb{R}$  satisfies three broad types of properties:

- **Algebraic Properties** – how numbers combine via addition and multiplication
- **Order Properties** – the natural order on the number line
- **Completeness Property** – ensures no “gaps” on the number line

This section focuses on the **algebraic properties**.

# Introduction to Group Theory

## What is Group Theory?

- A branch of abstract algebra.
- Studies algebraic structures called **groups**.
- Provides a framework to understand **symmetry**.

## Definition of a Group

- A set  $G$  with a binary operation  $(\circ)$  satisfying:
  - 1 **Closure:** If  $a, b \in G \Rightarrow a \circ b \in G$
  - 2 **Associativity:**  $(a \circ b) \circ c = a \circ (b \circ c)$
  - 3 **Identity:**  $\exists e \in G$  such that  $a \circ e = e \circ a = a$
  - 4 **Inverse:**  $\forall a \in G, \exists a^{-1} \in G$  such that  $a \circ a^{-1} = e$

# Examples and Applications

## Examples of Groups

- $(\mathbb{Z}, +)$ : Integers under addition.
- $(\mathbb{R} \setminus \{0\}, \times)$ : Non-zero reals under multiplication.
- Symmetry groups: Rotations and reflections of a square.

## Applications

- Mathematics: Number theory, geometry.
- Physics: Quantum mechanics, crystallography.
- Computer Science: Cryptography, coding theory.

# Algebraic Properties (I)

The set  $\mathbb{R}$  is equipped with two operations: addition (+) and multiplication ( $\cdot$ ).

## A1 Associativity:

$$a + (b + c) = (a + b) + c \quad \text{and} \quad a(bc) = (ab)c$$

## A2 Commutativity:

$$a + b = b + a \quad \text{and} \quad ab = ba$$

## A3 Identity Elements: There exist distinct $0, 1 \in \mathbb{R}$ such that:

$$a + 0 = a, \quad a \cdot 1 = a$$

# Algebraic Properties (II)

## A4 Existence of Inverses:

- For every  $a \in \mathbb{R}$ , there exists  $a'$  (additive inverse) such that:

$$a + a' = 0$$

- For every  $a \neq 0$ , there exists  $a^*$  (multiplicative inverse) such that:

$$a \cdot a^* = 1$$

## A5 Distributivity:

$$a(b + c) = ab + ac \quad \text{for all } a, b, c \in \mathbb{R}$$

# Consequences of Axioms

From properties A1–A5, we can deduce:

- $a \cdot 0 = 0$  for all  $a \in \mathbb{R}$
- The additive inverse is unique (called the **negative** of  $a$ )
- The multiplicative inverse is unique (called the **reciprocal** of  $a$ )
- Simplifications like:

$$a + (-b) = a - b, \quad \frac{1}{b} \text{ for } b \neq 0$$



# Simplified Expressions and Notation

Once the formal properties are understood, we use simplified expressions:

$$a(1/b), \quad a + a, \quad aa, \quad (a + b) + c, \quad (ab)c, \quad a + (-b)$$

**become:**

$$a/b, \quad 2a, \quad a^2, \quad a + b + c, \quad abc, \quad a - b$$

## Summation and Product Notation:

- For real numbers  $a_1, \dots, a_n$ , the sum:

$$a_1 + \cdots + a_n = \sum_{i=1}^n a_i$$

- The product:

$$a_1 \cdots a_n = \prod_{i=1}^n a_i$$

- Empty sum:  $\sum_{i=1}^0 a_i := 0$   
Empty product:  $\prod_{i=1}^0 a_i := 1$

# Example of a Non-Associative Binary Operation

**Define:** Let  $*$  be a binary operation on  $\mathbb{R}$  defined by:

$$a * b = a - b$$

**Check associativity:**

Let  $a = 5$ ,  $b = 3$ ,  $c = 2$ .

$$(a * b) * c = (5 - 3) - 2 = 2 - 2 = 0$$

$$a * (b * c) = 5 - (3 - 2) = 5 - 1 = 4$$

$$\Rightarrow (a * b) * c \neq a * (b * c)$$

**Conclusion:** Subtraction is **not associative**.

# Binary Operation: Not Distributive

**Define** a binary operation  $*$  on  $\mathbb{Z}$  as:

$$a * b = a + b + 1$$

**Check for distributivity:**

Let's test with **specific values** to see if distributivity holds.

**Example:** Let  $a = 1$ ,  $b = 2$ ,  $c = 3$ .

**Compute Left Side:**  $a * (b + c)$

$$b + c = 2 + 3 = 5$$

$$a * (b + c) = 1 * 5 = 1 + 5 + 1 = 7$$

**Compute Right Side:**  $(a * b) + (a * c)$

$$a * b = 1 * 2 = 1 + 2 + 1 = 4$$

$$a * c = 1 * 3 = 1 + 3 + 1 = 5$$

$$(a * b) + (a * c) = 4 + 5 = 9$$

They are **not equal**, so the operation is **not distributive**.

# Order Properties of $\mathbb{R}$

The set  $\mathbb{R}$  contains a subset  $\mathbb{R}^+$  (positive real numbers), satisfying:

O1

Given any  $a \in \mathbb{R}$ , exactly one of the following holds:

$$a \in \mathbb{R}^+, \quad a = 0, \quad -a \in \mathbb{R}^+$$

O2

If  $a, b \in \mathbb{R}^+$ , then:

$$a + b \in \mathbb{R}^+ \quad \text{and} \quad ab \in \mathbb{R}^+$$

# Defining the Order Relation

Define  $a < b$  for  $a, b \in \mathbb{R}$  if  $b - a \in \mathbb{R}^+$ . Then:

- $b > a$  means  $a < b$
- $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$

# Consequences of Order Properties

- ① Given  $a, b \in \mathbb{R}$ , exactly one of the following is true:

$$a < b, \quad a = b, \quad b < a$$

- ② If  $a < b$  and  $b < c$ , then  $a < c$

- ③ If  $a < b$ , then:

- $a + c < b + c$
- If  $c > 0$ , then  $ac < bc$
- If  $c < 0$ , then  $ac > bc$

# Notation

- $a \leq b$  means  $a < b$  or  $a = b$
- $a \geq b$  means  $a > b$  or  $a = b$

**Key result:**  $1 > 0$ , hence  $a^2 > 0$  for all  $a \in \mathbb{R}, a \neq 0$



# Supremum and Infimum of a Set

Let  $S \subseteq \mathbb{R}$ . Then:

- A real number  $M$  is the **supremum** of  $S$  if:
  - $M$  is an upper bound of  $S$
  - For every  $\epsilon > 0$ , there exists  $a \in S$  such that  $M - \epsilon < a \leq M$
- A real number  $m$  is the **infimum** of  $S$  if:
  - $m$  is a lower bound of  $S$
  - For every  $\epsilon > 0$ , there exists  $b \in S$  such that  $m \leq b < m + \epsilon$

# Illustration of Supremum and Infimum



**Fig. 1.2.** Illustration of the supremum  $M$  and the infimum  $m$  of a subset  $S$  of  $\mathbb{R}$ .

Figure:

# Examples and Notes

- Example: Let  $S = \{x \in \mathbb{R} : 0 < x \leq 1\}$

$$\inf S = 0, \quad \sup S = 1$$

- If  $\sup S \in S$ , then  $\max S = \sup S$
- If  $\inf S \in S$ , then  $\min S = \inf S$
- The empty set  $\emptyset$  has neither a supremum nor an infimum

**Thank You!**